Visual Question Answering

Q:All mandibular anterior teeth are visible in the lateral intraoral photograph?

A: True

B: False

Qwen3-4b Response:

The description notes that the mandibular anterior teeth are not fully visible, indicating that not all of them are captured in the image.

Lingshu-32b Response:

The overall description states that the image shows the maxillary and mandibular anterior teeth, indicating that all mandibular anterior teeth are present.

Dental Caries Detection

Q:Is dental caries visible in this image?

A: Yes

B: No

Qwen3-4b Response:

Yes

Lingshu-32b Response:

No

Tooth Counting

Q:Count the total number of visible teeth in this periapical radiograph.

Ground truth 2

Qwen3-4b Response:

2

Lingshu-32b Response:

6

Bone Resorption Grading

Prompt:

Grade the periodontal bone resorption severity based on this panoramic radiograph. Use the panoramic image together with the provided patient metadata. Return only one label: 1, 2, or 3.

Patient Metadata:

Age: 33, Gender: female, Missing Teeth: 4, Implant: 0, Residual Root: 0, Functional Tooth Count: 12

Qwen3-4b Response:

Grade 2

Lingshu-32b Response:

Grade 3

Clinical Image Description

Prompt: Generate a detailed and vivid natural language description based on this clinical image. The description should clearly describe the imaging direction, the main subject of the image, and all observed abnormalities.

Qwen3-4b Response:

The photograph presents a clear, side- view intraoral image of the patient's maxillary and mandibular arches, captured with the patient's mouth slightly open and the lips retracted using a mouth prop...

Lingshu-32b Response:

The photograph presents a clear view of the maxillary dentition taken from the right side, with the patient's mouth opened wide enough to expose the full complement of upper teeth. A metal lip retractor is positioned on the right side...

Structured Diagnostic Report

Prompt:Generate a structured diagnostic report based on the clinical images and patient information. The report should include: Patient Record, Examination, Radiographic Examination, Diagnosis, Treatment Plan, Treatment Recommendations, and Management.

Groundtruth: Patient Record: Follow-up consultation.
Diagnosis: Pulpitis (K04.000).
Examination: Tooth 35 shows no percussion pain or gingival inflammation and is temporarily sealed.
Diagnosis: Pulpitis in tooth 35 (K04.000).

Qwen3-4b Response:

Patient Record: Follow-up consultation. Diagnosis: Pulpitis (K04.005).
Examination: Tooth 36 is temporarily sealed, with no percussion pain or gingival inflammation.
Diagnosis: Pulpitis in tooth 36 (K04.005).

Lingshu-32b Response:

Patient Record: Follow-up consultation. Diagnosis: Pulpitis (K04.005).
Examination: Tooth 37 shows no percussion pain or gingival inflammation, with a temporary crown in place.
Diagnosis: Pulpitis in tooth 37 (K04.005).